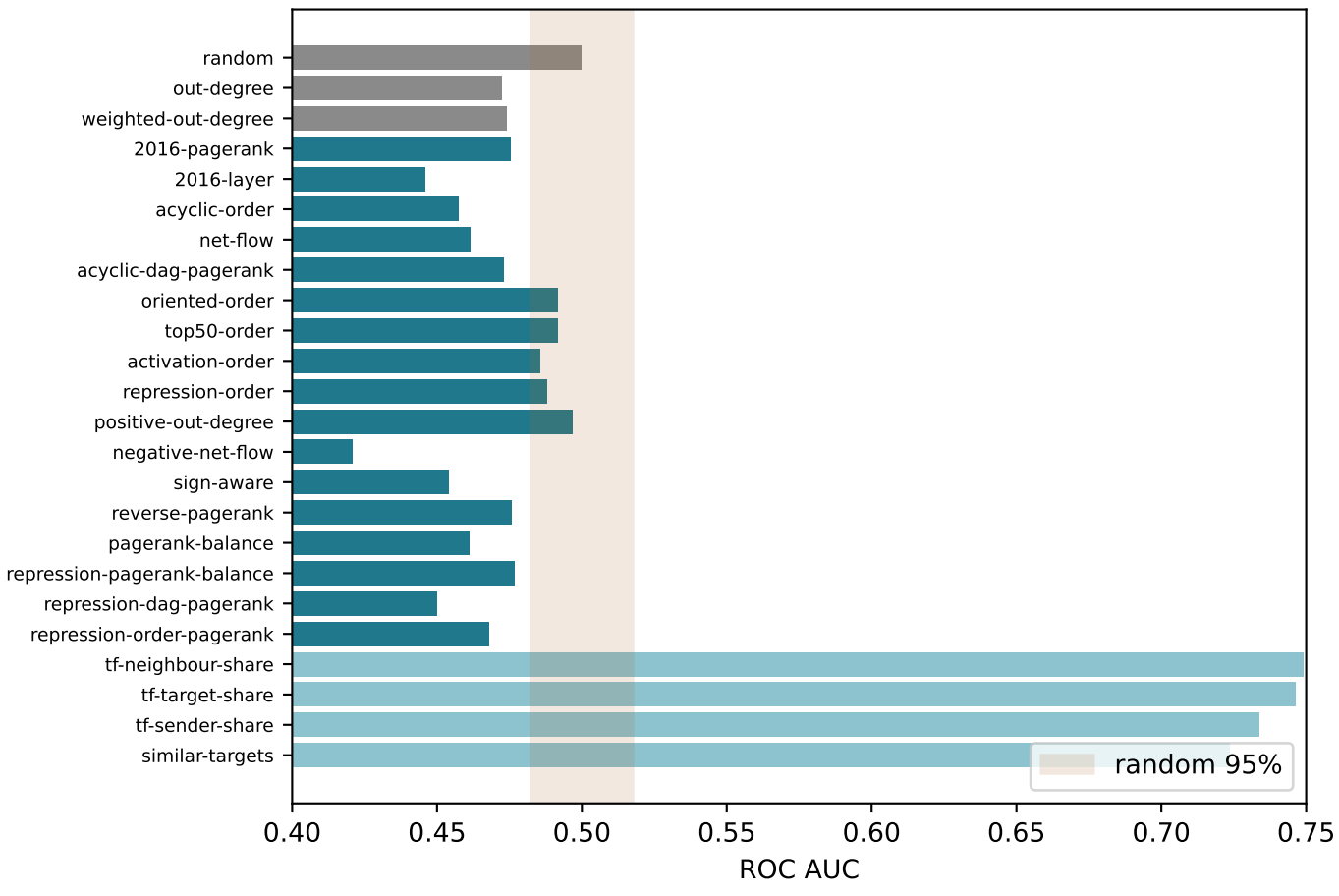
# Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,914 genes ranked, 1,147 in lambert-2018-tfs.txt · top k = 100

| Pipeline | Reference genes in top 100 | AUC | p top-k | p AUC | AUC vs best baseline (95%) | p vs baseline |
|---|---|---|---|---|---|---|
| random | 9.6 (4–16) | 0.500 (0.482–0.518) | | | | |
| out-degree | 1 | 0.472 | 1.000 | 1.000 | -0.002 (-0.003, -0.001) | 0.004 |
| weighted-out-degree | 2 | 0.474 | 1.000 | 0.997 | +0.000 (+0.000, +0.000) | |
| 2016-pagerank | 2 | 0.475 | 1.000 | 0.997 | +0.001 (-0.018, +0.021) | 0.908 |
| 2016-layer | 7 | 0.446 | 0.856 | 1.000 | -0.028 (-0.047, -0.007) | 0.010 |
| acyclic-order | 1 | 0.457 | 1.000 | 1.000 | -0.017 (-0.025, -0.009) | < 0.001 |
| net-flow | 1 | 0.462 | 1.000 | 1.000 | -0.013 (-0.027, +0.001) | 0.080 |
| acyclic-dag-pagerank | 1 | 0.473 | 1.000 | 1.000 | -0.001 (-0.008, +0.006) | 0.732 |
| oriented-order | 2 | 0.492 | 1.000 | 0.809 | +0.018 (-0.001, +0.037) | 0.064 |
| top50-order | 9 | 0.492 | 0.646 | 0.814 | +0.018 (-0.002, +0.036) | 0.076 |
| activation-order | 5 | 0.485 | 0.968 | 0.943 | +0.011 (+0.003, +0.020) | 0.012 |
| repression-order | 10 | 0.488 | 0.506 | 0.904 | +0.014 (-0.004, +0.031) | 0.126 |
| positive-out-degree † | 9 | 0.497 | 0.646 | 0.639 | +0.023 (+0.019, +0.027) | < 0.001 |
| negative-net-flow † | 0 | 0.421 | 1.000 | 1.000 | -0.053 (-0.070, -0.037) | < 0.001 |
| sign-aware † | 1 | 0.454 | 1.000 | 1.000 | -0.020 (-0.028, -0.012) | < 0.001 |
| reverse-pagerank † | 1 | 0.476 | 1.000 | 0.996 | +0.002 (-0.004, +0.007) | 0.546 |
| pagerank-balance † | 0 | 0.461 | 1.000 | 1.000 | -0.013 (-0.029, +0.003) | 0.130 |
| repression-pagerank-balance † | 5 | 0.477 | 0.968 | 0.995 | +0.003 (-0.014, +0.019) | 0.714 |
| repression-dag-pagerank † | 5 | 0.450 | 0.968 | 1.000 | -0.024 (-0.040, -0.008) | 0.006 |
| repression-order-pagerank † | 1 | 0.468 | 1.000 | 1.000 | -0.006 (-0.022, +0.010) | 0.424 |
| tf-neighbour-share ‡ | 96 | 0.749 | < 0.001 | < 0.001 | +0.275 (+0.253, +0.297) | < 0.001 |
| tf-target-share ‡ | 89 | 0.746 | < 0.001 | < 0.001 | +0.272 (+0.250, +0.294) | < 0.001 |
| tf-sender-share ‡ | 95 | 0.734 | < 0.001 | < 0.001 | +0.260 (+0.238, +0.283) | < 0.001 |
| similar-targets ‡ | 92 | 0.724 | < 0.001 | < 0.001 | +0.250 (+0.226, +0.273) | < 0.001 |

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



# Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

## collectri

| Graph | Pairs | Recall (×) | Positive × | Negative × | Sign (chance) |
|---|---|---|---|---|---|
| raw graph | 31,763 | 0.032 (2.85×) | 3.58× | 0.55× | 0.83 (0.82) |
| resolve_reciprocals → drop_sinks → prune_triangles | 31,744 | 0.002 (0.87×) | 2.68× | 0.58× | 0.48 (0.46) |
| resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering | 19,008 | 0.002 (0.76×) | 1.70× | 0.69× | 0.29 (0.29) |
| absolute | 31,763 | 0.032 (2.85×) | | | |
| absolute → minimum_feedback_order | 31,721 | 0.022 (2.65×) | | | |
| orient_reciprocals ratio=2.0 → absolute | 31,758 | 0.012 (1.82×) | | | |
| orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order | 31,758 | 0.012 (1.82×) | | | |
| top_outgoing k=50 → absolute | 31,763 | 0.014 (4.23×) | | | |
| top_outgoing k=50 → absolute → minimum_feedback_order | 31,713 | 0.009 (3.78×) | | | |
| keep_sign sign=1 → absolute | 31,756 | 0.030 (3.57×) | | | |
| keep_sign sign=1 → absolute → minimum_feedback_order | 31,636 | 0.020 (3.27×) | | | |
| keep_sign sign=-1 → absolute | 30,040 | 0.002 (0.54×) | | | |
| keep_sign sign=-1 → absolute → minimum_feedback_order | 26,368 | 0.001 (0.58×) | | | |
| keep_sign sign=-1 | 30,040 | 0.002 (0.54×) | | 0.54× | 0.24 (0.24) |

# Pipeline steps

### out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 1,828,303 edges (534,217 negative), largest strong component 11,909.

### weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 1,828,303 edges (534,217 negative), largest strong component 11,909.

### 2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 528,369 edges (479,033 negative), largest strong component 11,550.

### 2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 528,369 edges (479,033 negative), largest strong component 11,550; DAG keeps 303,499 edges.

### acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,828,303 edges (0 negative), largest strong component 11,909; DAG keeps 1,399,798 edges.

### net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 1,828,303 edges (0 negative), largest strong component 11,909.

### acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 1,828,303 edges (0 negative), largest strong component 11,909; DAG keeps 1,399,798 edges.

### oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,151,515 edges (0 negative), largest strong component 8,591; DAG keeps 1,141,284 edges.

### top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 493,328 edges (0 negative), largest strong component 11,886; DAG keeps 359,221 edges.

### activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,294,086 edges (0 negative), largest strong component 11,872; DAG keeps 946,278 edges.

### repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 534,217 edges (0 negative), largest strong component 10,674; DAG keeps 464,661 edges.

### positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 1,828,303 edges (534,217 negative), largest strong component 11,909.

### negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 1,828,303 edges (534,217 negative), largest strong component 11,909.

### sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 1,828,303 edges (534,217 negative), largest strong component 11,909.

### reverse-pagerank: Highest reverse PageRank (raw graph)